THE STRONGLY CARTESIAN SUBCATEGORY OF A FIBRED CATEGORY IS FIBRED IN GROUPOIDS

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Editorial note. Counted span is the author's own proof environment, source lines 7334-7359, following the statement at 7317-7332.

Editorial note. The second lifting property leans on the in-chapter discussion following definition-cartesian-over-C; the invoked steps are stated in-span.

Allowed standard prerequisites (editorial).

- Closure of strongly cartesian morphisms. Lemma lemma-composition-cartesian of this chapter, invoked at proof lines 7335-7338 and 7357-7358 (in-chapter input; identity is strongly cartesian and composition closes). Makes the subcategory construction well defined and supplies that isomorphisms are strongly cartesian.
- Definition of fibred in groupoids and following discussion. Definition definition-fibred-groupoids of this chapter with the discussion and equation-fibred-groupoids following it, used at 7339-7356. The two lifting properties; the second is verified via the same-diagram argument referenced from the discussion.
- Uniqueness of strongly cartesian morphisms over the same arrow. Discussion following definition-cartesian-over-C of this chapter, invoked at 7354-7356. Two strongly cartesian morphisms over the same base arrow with the same target are isomorphic.

Source reference. Original definition 02XK: sources/stacks-categories/pinned-original-source.tex, source lines 6422–6422. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 003T: sources/stacks-categories/pinned-original-source.tex, source lines 7195–7195. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 03WP: sources/stacks-categories/pinned-original-source.tex, source lines 7227–7227. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

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Source reference. Original lemma 02XL: sources/stacks-categories/pinned-original-source.tex, source lines 6459–6459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let $p : \mathcal{S} \to \mathcal{C}$ be a fibred category. Let $\mathcal{S}'$ be the subcategory of $\mathcal{S}$ defined as follows*

(1) $\mathrm{Ob}(\mathcal{S}') = \mathrm{Ob}(\mathcal{S})$, *and*
(2) *for $x, y \in \mathrm{Ob}(\mathcal{S}')$ the set of morphisms between x and y in $\mathcal{S}'$ is the set of strongly cartesian morphisms between x and y in $\mathcal{S}$.*

Let $p' : \mathcal{S}' \to \mathcal{C}$ be the restriction of p to $\mathcal{S}'$. Then $p' : \mathcal{S}' \to \mathcal{C}$ is fibred in groupoids.

Proof. Note that the construction makes sense since by Lemma 02XL the identity morphism of any object of $\mathcal{S}$ is strongly cartesian, and the composition of strongly cartesian morphisms is strongly cartesian. The first lifting property of Definition 003T follows from the condition that in a fibred category given any morphism $f : V \to U$ and x lying over U there exists a strongly cartesian morphism $\varphi : y \to x$ lying over f. Let us check the second lifting property of Definition 003T for the category $p' : \mathcal{S}' \to \mathcal{C}$ over $\mathcal{C}$. To do this we argue as in the discussion following Definition 003T. Thus in Diagram 03WP the morphisms ϕ, ψ and γ are strongly cartesian morphisms of $\mathcal{S}$. Hence γ and $\phi \circ \psi$ are strongly cartesian morphisms of $\mathcal{S}$ lying over the same arrow of $\mathcal{C}$ and having the same target in $\mathcal{S}$. By the discussion following Definition 02XK this means these two arrows are isomorphic as desired (here we use also that any isomorphism in $\mathcal{S}$ is strongly cartesian, by Lemma 02XL again). $\square$